SEAA 4333 REINFORCED CONCRETE DESIGN II · WEEK 7

Columns I

Braced or unbraced, effective length, short or slender

Workbook ch. 7 | ABCD card | Phone or laptop

Column Lab

Scan for this page’s Explorer tab. You will use it in three activities today.

TODAY · 180 MINUTES

How the next three hours run

0:00 **Warm-up** Units 3–6 in three votes	0:10 **Hook** The ruler	0:15 **Columns, l_0** Braced or unbraced?	0:35 **Predict, reveal** Lab tab 1
0:45 **k, λ, λlim** The EC2 check	1:05 **Faded example** In pairs	1:30 **Break** 10 minutes	1:40 **Concept checks** Which axis? Which sign?
1:47 **The race** Eight columns, teams	2:05 **Exam-size problem** Alone, 30 minutes	2:35 **Studio, exit** Your column	

No talk longer than ten minutes. Rust times are your turn to work.

WARM-UP · NO NOTES · CARDS UP

Three questions from the frame analysis

W1 · Unit 6

Portal method: in a sway frame under wind, where is the moment zero in each column?

A at the base

B at mid-height

C at the top

D nowhere

W2 · Unit 5

Cantilever method: the axial forces in the columns of a storey are proportional to

A their areas

B their stiffnesses

C their distances from the centroid of the column group

D the bay widths

W3 · Unit 3

One-free-joint sub-frame for column moments: the beams enter with

A half their stiffness, 0.5I/L

B their full stiffness, I/L

C twice their stiffness

D zero stiffness

WARM-UP · ANSWERS

B, C, A, and what each means for a column

W1

B: mid-height

An unbraced column bends in double curvature, with a point of contraflexure in its height.

W2

C: distance from the centroid

Edge columns carry the largest wind axial force; it adds to N_{Ed} in the column design.

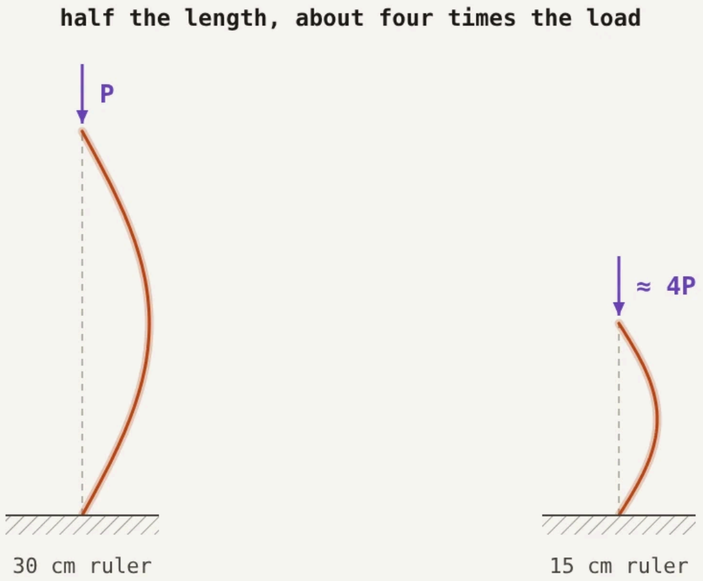
W3

A: half stiffness

Today EC2 counts each beam as 2EI/l in k: a cracked stiffness, used for buckling.

HOOK · 5 MINUTES

Why does a long ruler buckle when a short one does not?



In pairs: press a ruler end to end until it bows. Then hold it at mid-length and press again.

Write one sentence: what changed, and why?

$$N_{cr} = \pi^2 EI / l_0^2$$

BLOCK A · WHERE THIS SITS

Three weeks of columns; today is step 6 of 9

Unit 7 · today

Classify

Braced or unbraced; l_0 from the beams; λ against λlim about each axis.

Unit 8 · week 9

Short columns

Axial load and moment; the N–M interaction; bars and links.

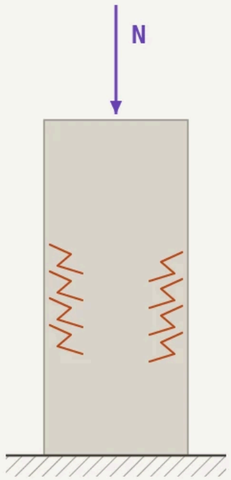
Unit 9 · week 10

Slender and biaxial

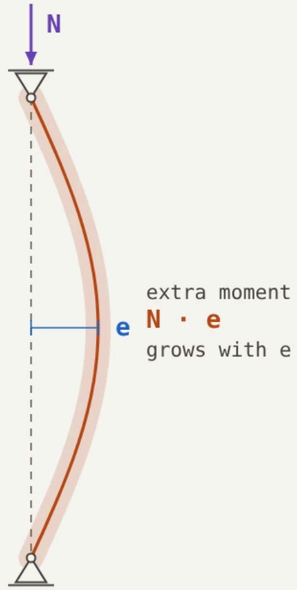
The nominal curvature moment M_2; bending about both axes.

Steps 1–5 (exposure, materials, size, cover, N and M) you know from RCD I and Units 3–6. Today: step 6, short or slender. Then the design moment, the bars and the detailing.

Short columns crush; slender ones bend first



short: the section crushes



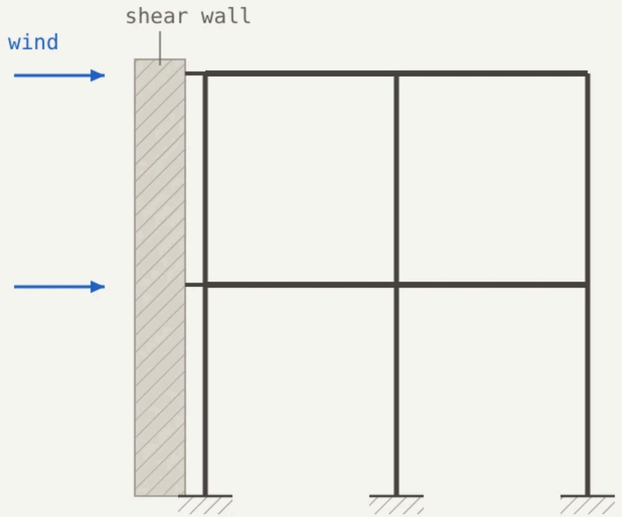
slender: it bends sideways first

A column (EC2 5.3.1(7)): h ≤ 4b and a height of at least 3h. Otherwise a wall.

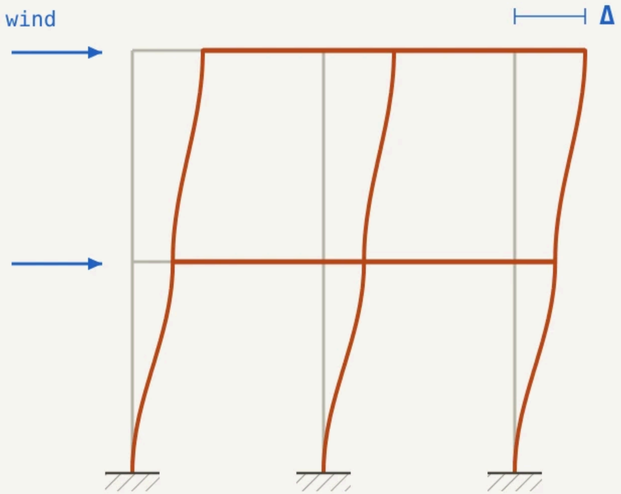
It carries N and moments: continuity, sway, and a minimum eccentricity.

Second-order effect: N times the deflection. EC2 ignores it only below λlim.

Braced or unbraced? Ask who carries the wind



braced: the wall takes the wind



unbraced: the frame sways by Δ

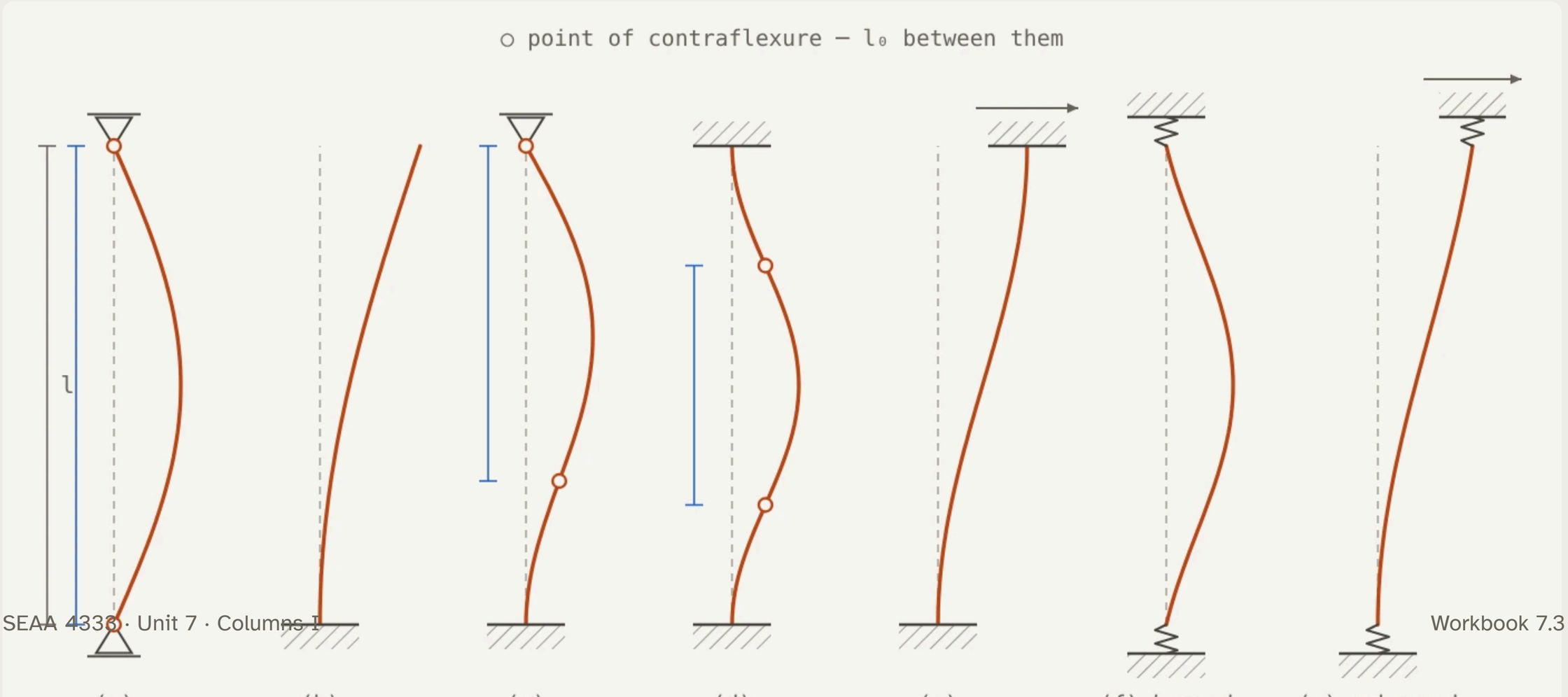
Braced: walls or a core carry the horizontal load; the column does not sway.

Unbraced: the frame bends to resist it; the column sways.

Our project frame has no walls: its columns are unbraced.

BLOCK A · TALK 1 OF 2

Effective length: the pin-ended column that buckles at the same load



CONCEPT CHECK 1 · THINK 60 S, THEN VOTE

Four columns, same section and clear height I. Which has the lowest buckling load?

A Pinned at both ends

B Fixed at the base, free at the top

C Fixed at the base, pinned at the top

D Fixed at both ends

CONCEPT CHECK 1 · ANSWER

B: the free top sways, so $l_0 = 2l$

A

$l_0 = l$

N_{cr} as Euler

B

$l_0 = 2l$

$N_{cr} \div 4$: the lowest

C

$l_0 = 0.7l$

$N_{cr} \times 2.0$

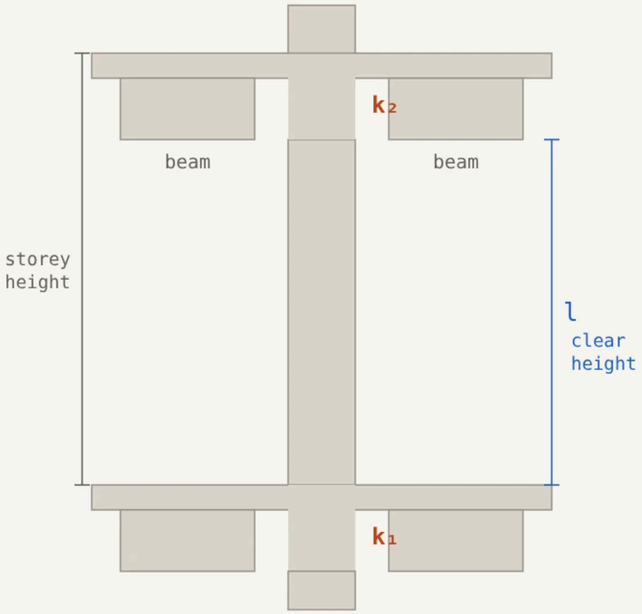
D

$l_0 = 0.5l$

$N_{cr} \times 4$

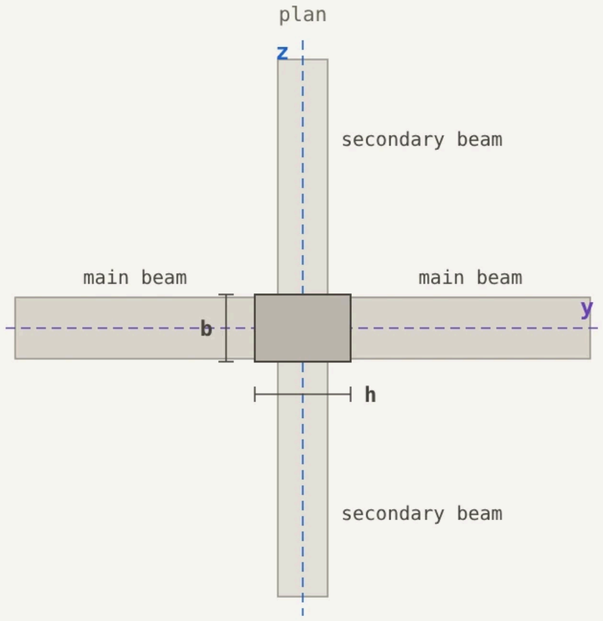
From the weakest to the strongest the buckling loads differ sixteen-fold, for the same column. The ends decide.

In a frame, the beams set k at each end



k = (I/l)col / Σ 2(I/l)beam ≥ 0.1

at each end: k_2 at the top, k_1 at the bottom



about z: $I = bh^3/12$, the main beams

about y: $I = hb^3/12$, the secondary beams

Braced (5.15)

$$l_0 = 0.5l\sqrt{[(1 + k_1/(0.45 + k_1))(1 + k_2/(0.45 + k_2))]}$$

Unbraced (5.16)

$$l_0 = l \cdot \max\{\sqrt{(1 + 10k_1k_2/(k_1 + k_2))};\ (1 + k_1/(1 + k_1))(1 + k_2/(1 + k_2))\}$$

$k \geq 0.1$. Fixed base: $k_1 = 0.1$. Pinned end: $k = \infty$.

ACTIVITY 7A · PREDICT, THEN REVEAL IN LAB TAB 1

A braced column between stiff beams (k = 0.1 at both ends) has $l_o = 0.59l$. The walls are taken away and the same column is now unbraced. Predict l_o.

A 0.59l: nothing changes

B 0.85l

C 1.22l

D 2.0l

Vote · convince your neighbour · vote again

ACTIVITY 7A · REVEAL

C: 1.22l, about twice as long, and λlim falls too

Braced

$l_0 = 0.5l \times (1 + 0.1/0.55) = 0.59l$

C from the end moments: up to 2.7

Unbraced

$\sqrt{(1 + 10 \times 0.01/0.2)} = 1.22$

$(1 + 0.1/1.1)^2 = 1.19$

C = 0.7 whatever the moments

So

λ doubles and λlim drops: an unbraced column is almost always slender.

BLOCK B · TALK

Slenderness: λ = l_0/i, about each axis

i = √(I/A)

radius of gyration of the uncracked concrete section

i = h/√12 = 0.289h

rectangle; h is the side in the plane of bending

λ ≈ 3.5 l_0/h

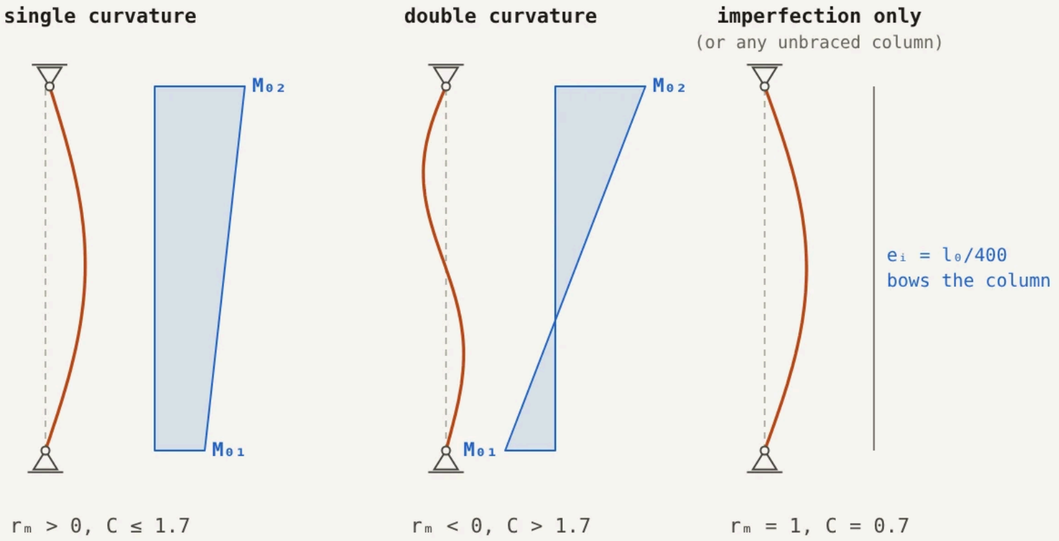
estimate it first, then calculate

Check both axes: each has its own l_0 (its own beams and clear height) and its own i. The minor axis usually governs, but not always.

The notes also give BS 8110 β-factors (Tables 3.19, 3.20): a quick, usually longer, cross-check. Design with Eq. 5.15 and 5.16.

BLOCK B · TALK

The limit: λlim = 20ABC/√n



A = 0.7, B = 1.1 if unknown

n = NEd/(Ac fcd), fcd = 0.567fck

C = 1.7 - rm, rm = M01/M02

|M02| ≥ |M01|. Double curvature: rm < 0.

rm = 1 (C = 0.7) if unbraced, or if |M02| ≤ NEd l_0/400.

BLOCK B · WORKED EXAMPLE 7.1 · I DO

A braced column between stiff beams

Braced; floors 6.0 m apart

Column 250 × 300; C30/37

z: one main beam 250 × 600, 6.0 m

y: two secondary 250 × 600, 4.5 m

NEd 1200 kN

Mz 60 / 35 (opposite faces)

My 1 / 0 kN m

App case C1

Calculation	Output
l = 6000 - 600 (both axes)	5400 mm
k = 1.042/(2 × 7.50); 0.7234/(2 × 20.0)	0.069; 0.018 → 0.1
l_0 = 0.5 × 5400 × 1.182	3191 mm
λz = 3191/86.6; λy = 3191/72.2	36.8; 44.2
n = 1200 × 10^3/(75 000 × 17.0)	0.941
Cz: rm = -35/60; Cy: 1 ≤ 9.6 kN m	2.283; 0.7
λlim = 15.4 C/√0.941	36.2; 11.1
Verdict	**slender, both axes**

BLOCK B · THE CHECK

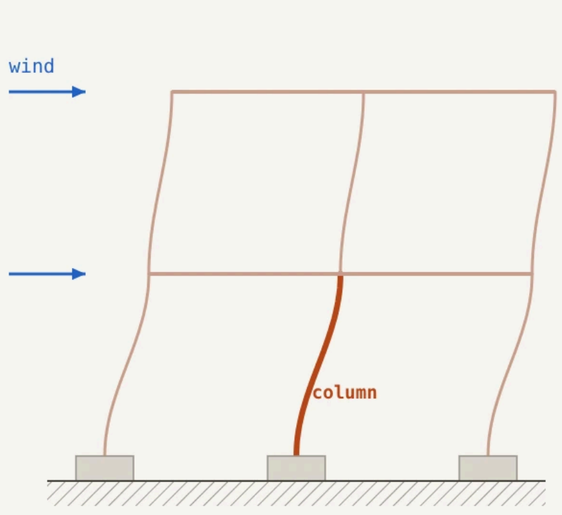
The slenderness check in eight lines

1. Braced or unbraced?
2. Clear height l, per axis
3. $K = I/l$; k at each end, ≥ 0.1
4. l_0 by Eq. 5.15 or 5.16
5. $i = h/\sqrt{12}$; $\lambda = l_0/i$
6. $n = N_{Ed}/(A_c f_{cd})$
7. r_m and C, with the imperfection check
8. $\lambda_{lim} = 20ABC/\sqrt{n}$; the verdict, per axis

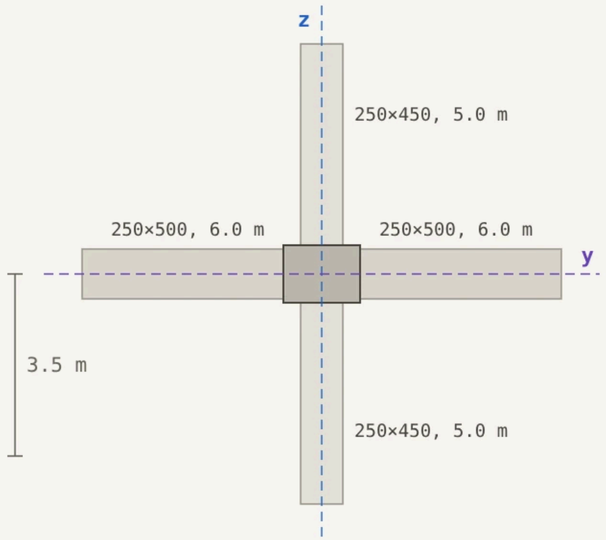
Rust steps are where most marks are lost: the bracing decision and the sign of r_m.

ACTIVITY 7B · PAIRS · 20 MINUTES

Faded Example 7.2: an unbraced column on a fixed base



unbraced frame: fixed bases, no walls



plan at the first floor: column 300 × 400

Steps 1–2 are done for you in Workbook 7.6. Finish steps 3–6 in pairs.

NEd 1600 kN; C30/37; z-beams 250 × 500 at 6.0 m both sides; y-beams 250 × 450 at 5.0 m both sides.

Then check in Lab tab 1 (case FE 7.2).

ACTIVITY 7B · CHECK

Slender about both axes; braced, it would not be

Calculation	Output
k_2: 5.333/(2 × 8.681); 2.951/(2 × 7.594)	0.3072; 0.1943
z: max(1.3245; 1.3473) × 3000	4042 mm
y: max(1.2885; 1.2684) × 3050	3930 mm
λz = 4042/115.5; λy = 3930/86.6	35.0; 45.4
n = 1600 × 10^3/(120 000 × 17.0)	0.784
λlim = 20 × 0.7 × 1.1 × 0.7/√0.784	12.2

Braced, the same column: l_0 = 1933 / 1891 mm, λ = 16.7 / 21.8, λlim ≈ 44: non-slender. Bracing changes the verdict, not the column.

1:30

Break

10 minutes. Back at 1:40.

Next: two quick votes, then a race between the teams.

CONCEPT CHECK 2 · THINK 60 S, THEN VOTE

A braced 250 × 500 column has l_0 = 3.0 m about both axes. Which is right?

A λ = 20.8 about both axes

B λ = 41.6 about the minor axis, 20.8 about the major

C λ = 41.6 about the major axis

D λ = 12.0 about the minor axis

CONCEPT CHECK 3 · THINK 60 S, THEN VOTE

Braced column: end moments 60 kN m (top) and 35 kN m (bottom), tension on opposite faces. NEd $l_0/400$ = 9.6 kN m. What is C?

A 1.12

B 2.28

C 0.70

D 1.70

CONCEPT CHECKS 2 AND 3 · ANSWERS

B and B

CC2

i = 250/√12 = 72.2 mm

λ = 3000/72.2 = 41.6

About the major axis i = 144.3 mm and λ = 20.8. The minor axis governs here.

CC3

rm = -35/60 = -0.583

C = 1.7 + 0.583 = 2.28

Taking rm positive gives 1.12: a limit half the size, and columns sent to Unit 9 for nothing.

ACTIVITY 7C · TEAMS OF FOUR · 18 MINUTES

The race: classify eight columns

1 Braced (core, walls). 300 × 300, l = 3.3 m, $k_1 = k_2 = 0.1$. NEd 1300, C30/37. M 40 / 20, opposite faces.

2 As card 1, but the moments 40 / 20 give tension on the same face.

3 As card 1, with no end moments (balanced spans).

4 Bare frame, ground floor, fixed base. 300 × 400, major axis, l = 3.0 m, $k_1 = 0.1$, $k_2 = 0.4$. NEd 1600, C30/37.

5 Braced, top storey. 300 × 300, l = 3.0 m, $k_1 = k_2 = 0.3$. NEd 250, C30/37. M 60 / 30, opposite faces.

6 Braced. 250 × 500, minor axis, l = 3.6 m, $k_1 = k_2 = 0.15$. NEd 1800, C30/37. M 8 / 4 about it, opposite faces.

7 Bare frame, top storey. 300 × 300, l = 3.0 m, $k_1 = k_2 = 0.5$. NEd 300, C30/37.

8 Braced. Pad not designed for moment: pinned base, $k_1 = \infty$; $k_2 = 0.2$. 300 × 300, l = 3.0 m. NEd 900, C25/30. M 30 top, 0 base.

For each card: braced or unbraced, l_0, λ, λlim, verdict. Two cards each, then check each other. Enter the team's answers in Lab tab 2.

ACTIVITY 7C · ANSWERS

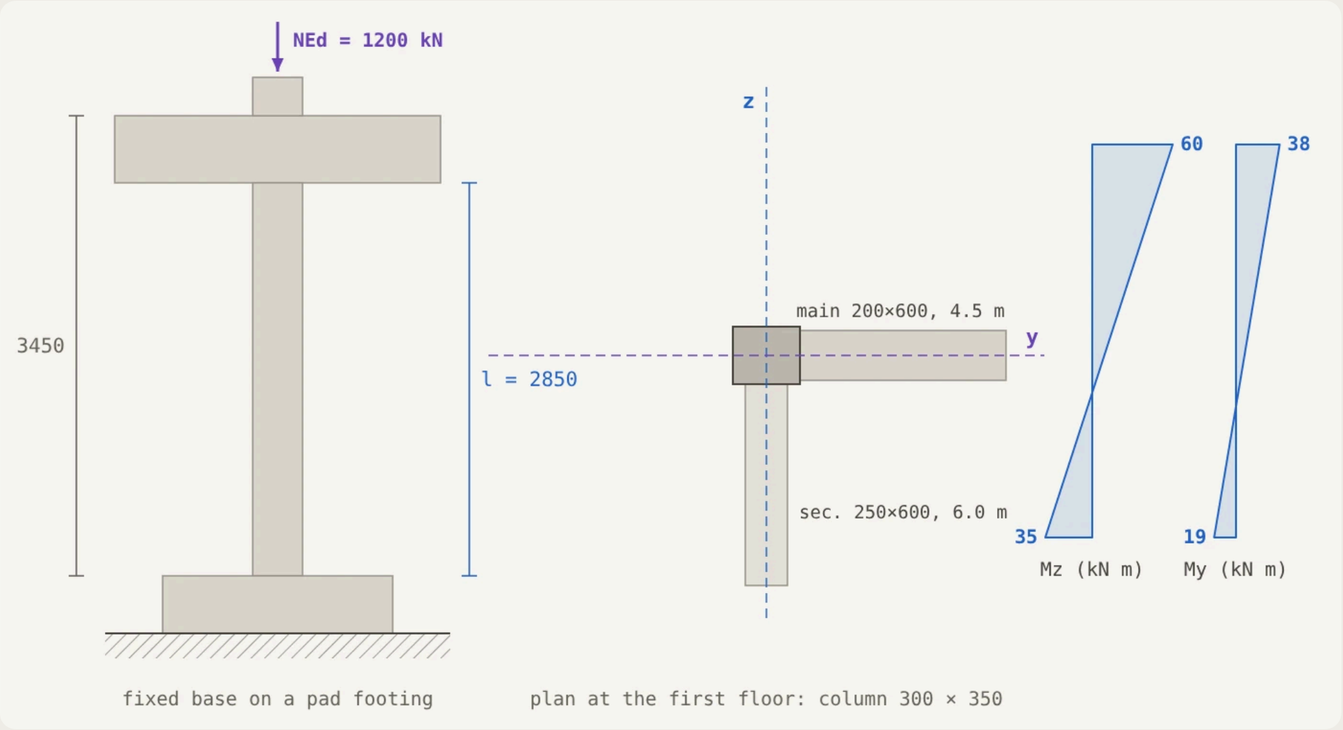
Five slender, three short

Card	l_0 (mm)	λ	λlim	Verdict
1	1950	22.5	36.8	non-slender
2	1950	22.5	20.0	slender
3	1950	22.5	11.7	slender
4	4208	36.4	12.2	slender
5	2100	24.2	83.8	non-slender
6	2250	31.2	11.7	slender
7	5612	64.8	24.3	slender
8	2426	28.0	31.2	non-slender

1 → 2 → 3: the same column, only the moments change. 6: 8 kN m is less than NEd $l_0/400$ = 10.1 kN m, so C = 0.7.

BLOCK D · ALONE · 30 MINUTES

D7: a past-paper question (15 marks)



Braced. Classify the column about both axes, in the Ref | Calculations | Output format.

C30/37. Beams 600 deep. Fixed base: k_1 = 0.1.

No app, no Lab, notes closed. Then self-mark with the answer scheme after class.

PROJECT STUDIO · 17 MINUTES

Is your column short or slender?

1

Your frame is unbraced: Eq. 5.16, C = 0.7.

2

Clear heights and k from your beam sizes, about both axes; fixed base k_1 = 0.1.

3

λ, λlim with your NEd from Unit 6; the verdict per axis.

Write the log entry in Workbook 7.10: the numbers and one decision (for example, a deeper beam or a bigger column).

PROFESSIONAL HABIT 7

Estimate λ before you calculate it

$\lambda \approx 3.5\ l_0/h$

a 300 mm column with l_0 = 3 m: λ ≈ 35

$0.59l \le l_0 \le l$

braced, with k ≥ 0.1

$l_0 \ge 1.22l$

unbraced, with k ≥ 0.1

A result outside these ranges is a slip: find it before you go on.

AFTER CLASS

Before Unit 8

Notes §1–§6 and the exercises on the Unit 7 page (45 min)

Column S7 in Lab tab 3: four slips to find

Self-mark D7 with the answer scheme on UTM eLearning

Bring N and M of your column, and read notes Examples 2 and 3

EXIT TICKET · 5 MINUTES · WORKBOOK 7.11

Before you leave

1 A braced 300 × 300 column, l = 3.0 m between stiff beams (k = 0.1 at both ends), NEd = 1500 kN, C30/37, no end moments. Slender or not? One line of working.

2 Why is an unbraced column almost always slender?

3 What is still unclear for you today?

Tear off the slip and hand it in at the door. Next week: Test 1, then short columns.